Part A: Basic Network Test Commands

1. ifconfig (Interface Configuration)

Protocol/Layer: It operates at the **Data Link Layer (Layer 2)** and **Network Layer (Layer 3)** configuration level. It does not use a specific transmission protocol to function but interacts with the **Operating System Kernel** to configure network interfaces (IP address, Subnet Mask, MAC address).

```
zeyun@DESKTOP-3D017KB:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.19.62.224  netmask 255.255.240.0  broadcast 172.19.63.255
        inet6 fe80::215:5dff:fe04:80dc  prefixlen 64  scopeid 0x20<link>
        ether 00:15:5d:04:80:dc  txqueuelen 1000  (Ethernet)
        RX packets 1679  bytes 763237 (763.2 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 669  bytes 63974 (63.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 107  bytes 8885 (8.8 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 107  bytes 8885 (8.8 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

2. ping (Packet Internet Groper)

Protocol: ICMP (Internet Control Message Protocol).

Mechanism: It sends **ICMP Echo Request** messages and waits for **ICMP Echo Reply** messages.

```
zeyun@DESKTOP-3D017KB:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=106 time=16.4 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=106 time=18.4 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=106 time=16.6 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=106 time=16.0 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=106 time=17.2 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=106 time=16.8 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=106 time=18.2 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=106 time=17.5 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=106 time=16.8 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=106 time=14.3 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=106 time=21.3 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=106 time=18.2 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=106 time=20.9 ms
^C
--- 8.8.8.8 ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12019ms
rtt min/avg/max/mdev = 14.334/17.594/21.345/1.828 ms
```

3. nslookup (Name Server Lookup)

Protocol: DNS (Domain Name System).

Mechanism: It typically uses **UDP**(Port 53) for standard queries, and **TCP** (Port 53) for larger responses or zone transfers occasionally.

```
zeyun@DESKTOP-3D017KB:~$ nslookup skyzhou.top
Server:         10.255.255.254
Address:        10.255.255.254#53

Non-authoritative answer:
Name:   skyzhou.top
Address: 104.21.81.43
Name:   skyzhou.top
Address: 172.67.138.27
Name:   skyzhou.top
Address: 2606:4700:3033::ac43:8a1b
Name:   skyzhou.top
Address: 2606:4700:3035::6815:512b
```

4. arp (Address Resolution Protocol)

Protocol: ARP (Address Resolution Protocol).

Mechanism: It maps a known **IP address** (Network Layer) to a **MAC address** (Data Link Layer).

```
zeyun@DESKTOP-3D017KB:~$ arp -a
DESKTOP-3D017KB.mshome.net (172.19.48.1) at 00:15:5d:04:85:cd [ether] on eth0
```

5. netstat (Network Statistics)

Protocol: It displays statistics for multiple protocols, primarily **TCP (Transmission Control Protocol)**, **UDP (User Datagram Protocol)**, **ICMP**, and **IP**.

Mechanism: It reads information directly from the **OS Kernel's network stack tables**.

```
zeyun@DESKTOP-3D017KB:~$ netstat -anop
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name     Timer
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      -                    off (0.00/0/0)
tcp        0      0 10.255.255.254:53       0.0.0.0:*               LISTEN      -                    off (0.00/0/0)
udp        0      0 127.0.0.53:53           0.0.0.0:*                           -                    off (0.00/0/0)
udp        0      0 10.255.255.254:53       0.0.0.0:*                           -                    off (0.00/0/0)
udp        0      0 127.0.0.1:323           0.0.0.0:*                           -                    off (0.00/0/0)
udp6       0      0 ::1:323                 :::*                                -                    off (0.00/0/0)
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   PID/Program name     Path
unix  3      [ ]         STREAM     CONNECTED     27202    -                    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     23004    -                    /tmp/dbus-pq9FZ3QlB7
unix  3      [ ]         DGRAM      CONNECTED     19896    -
unix  3      [ ]         STREAM     CONNECTED     24200    -                    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     11965    -
unix  2      [ ACC ]     STREAM     LISTENING     2748     -                    /run/WSL/2_interop
unix  2      [ ACC ]     STREAM     LISTENING     20624    -                    /run/WSL/1_interop
unix  3      [ ]         STREAM     CONNECTED     13162    -                    /mnt/wslg/PulseAudioRDPSink
unix  2      [ ACC ]     SEQPACKET  LISTENING     11837    -                    /mnt/wslg/weston-notify.sock
unix  2      [ ACC ]     STREAM     LISTENING     18804    -                    /var/run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     24199    -                    /run/dbus/system_bus_socket
unix  2      [ ACC ]     STREAM     LISTENING     25614    -                    /mnt/wslg/runtime-dir/wayland-0
unix  2      [ ACC ]     STREAM     LISTENING     25617    -                    /tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     11952    -
unix  3      [ ]         STREAM     CONNECTED     11969    -                    /run/dbus/system_bus_socket
unix  2      [ ]         DGRAM                    28922    634/systemd          /run/user/1000/systemd/notify
unix  2      [ ACC ]     STREAM     LISTENING     28925    634/systemd          /run/user/1000/systemd/private
unix  2      [ ACC ]     STREAM     LISTENING     28930    634/systemd          /run/user/1000/bus
```

6. traceroute (Trace Route)

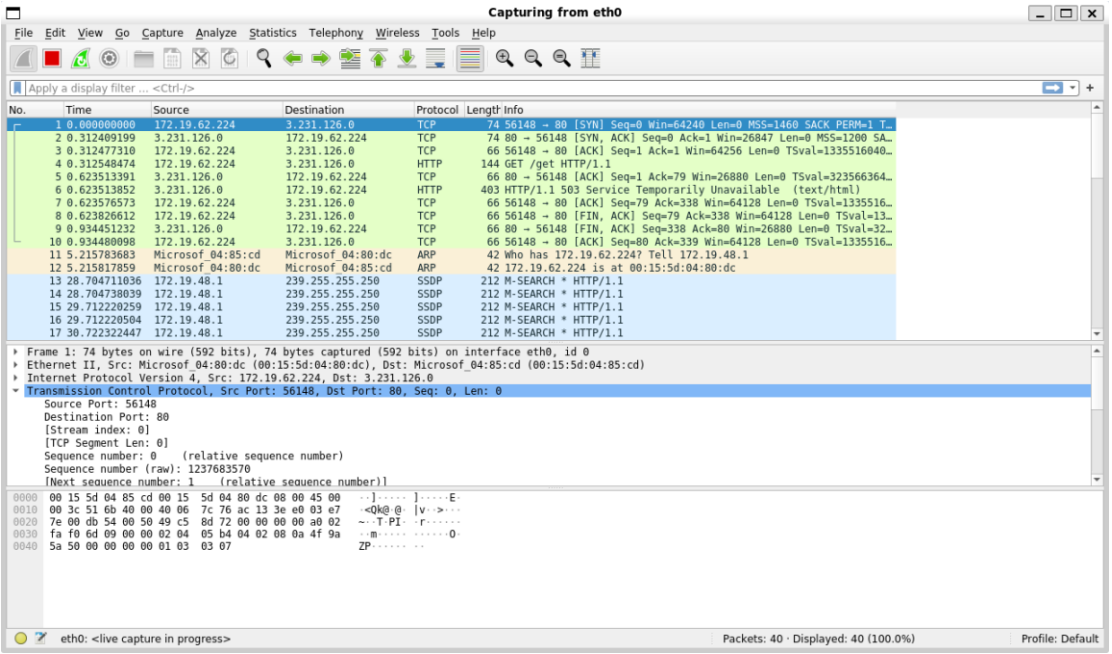
Protocol: By default, it uses **UDP** packets sent to high-numbered ports (33434+).

Mechanism: Rely on receiving **ICMP Time Exceeded** messages from intermediate routers to identify the path.

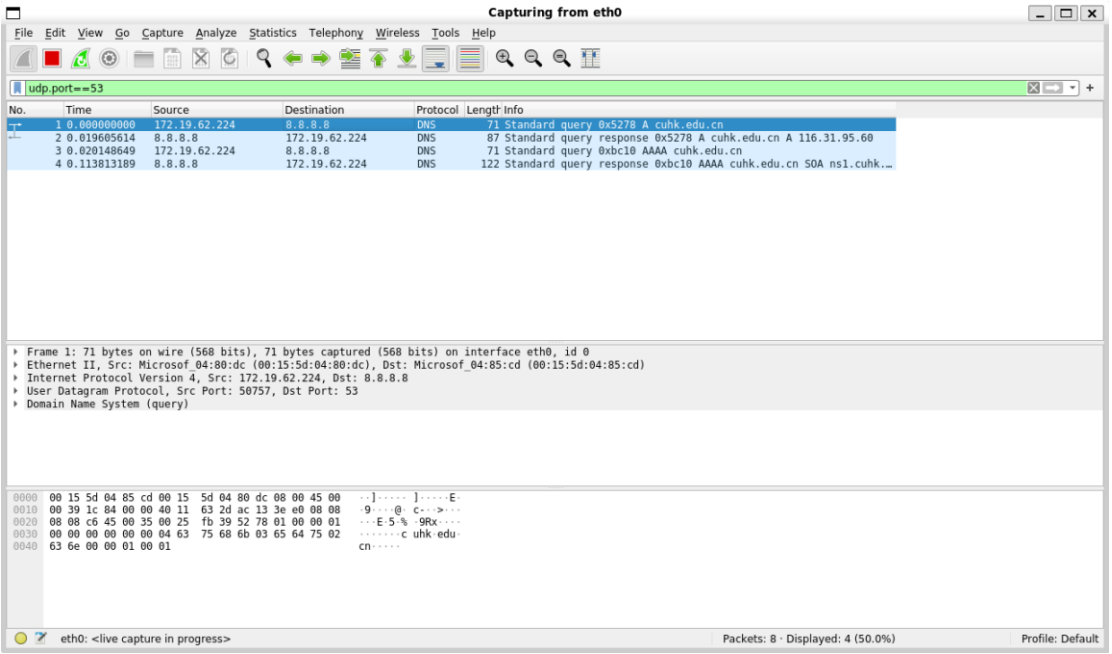
```
zeyun@DESKTOP-3D017KB:~$ traceroute 104.21.81.43
traceroute to 104.21.81.43 (104.21.81.43), 30 hops max, 60 byte packets
 1  DESKTOP-3D017KB.mshome.net (172.19.48.1)  0.467 ms  0.411 ms  0.372 ms
 2  10.31.15.254 (10.31.15.254)  59.219 ms  59.186 ms  59.157 ms
 3  10.40.0.53 (10.40.0.53)  7.517 ms  7.506 ms  7.496 ms
 4  10.20.238.25 (10.20.238.25)  5.367 ms  5.351 ms  5.340 ms
 5  10.20.238.1 (10.20.238.1)  5.386 ms  5.376 ms  5.346 ms
 6  10.20.238.19 (10.20.238.19)  5.256 ms  4.890 ms  4.875 ms
 7  116.31.95.1 (116.31.95.1)  4.864 ms *  9.561 ms
 8  59.37.186.189 (59.37.186.189)  7.181 ms 59.37.186.185 (59.37.186.185)  7.164 ms  7.154 ms
 9  14.147.126.86 (14.147.126.86)  9.518 ms 14.147.126.90 (14.147.126.90)  9.503 ms  7.543 ms
10  * 14.147.126.85 (14.147.126.85)  13.298 ms *
11  * * 14.147.127.5 (14.147.127.5)  14.199 ms
12  * * 202.97.71.238 (202.97.71.238)  14.513 ms
13  202.97.66.225 (202.97.66.225)  14.496 ms 202.97.12.17 (202.97.12.17)  14.486 ms 202.97.65.89 (202.97.65.89)  14.480 ms
14  202.97.107.162 (202.97.107.162)  195.926 ms  189.927 ms 202.97.116.250 (202.97.116.250)  187.191 ms
15  81.173.18.74 (81.173.18.74)  191.140 ms * 81.173.18.70 (81.173.18.70)  193.646 ms
16  141.101.65.93 (141.101.65.93)  185.615 ms 141.101.65.137 (141.101.65.137)  189.949 ms 141.101.65.145 (141.101.65.145)  189.944
ms
17  104.21.81.43 (104.21.81.43)  187.563 ms  187.549 ms  187.537 ms
```

Part B: Capture TCP/UDP Packets and Connection Process [15 points]

1. TCP Three-Way Handshake & Four-Way Handshake



2. UDP Data Transmission Analysis



3.Protocol Selection & Application Scenarios

1. Why is TCP chosen for HTTP?

Scenario: Web Browsing / File Transfer.

Reason: HTTP transfers text, images, and scripts where **correctness is mandatory**.

If a packet is lost during a file download, the file becomes corrupted.

TCP provides **Reliability** (Retransmission) and **Ordering** (Sequence Numbers) to ensure the webpage renders perfectly.

The overhead of the 3-way handshake is acceptable to guarantee data integrity.

2. Why is UDP chosen for DNS?

Scenario: Domain Name Resolution.

Reason: DNS queries are small and require **speed**.

A DNS query fits inside a single packet.

Using TCP would require 3 packets for handshake + 2 for data + 4 for teardown = **9 packets** total.

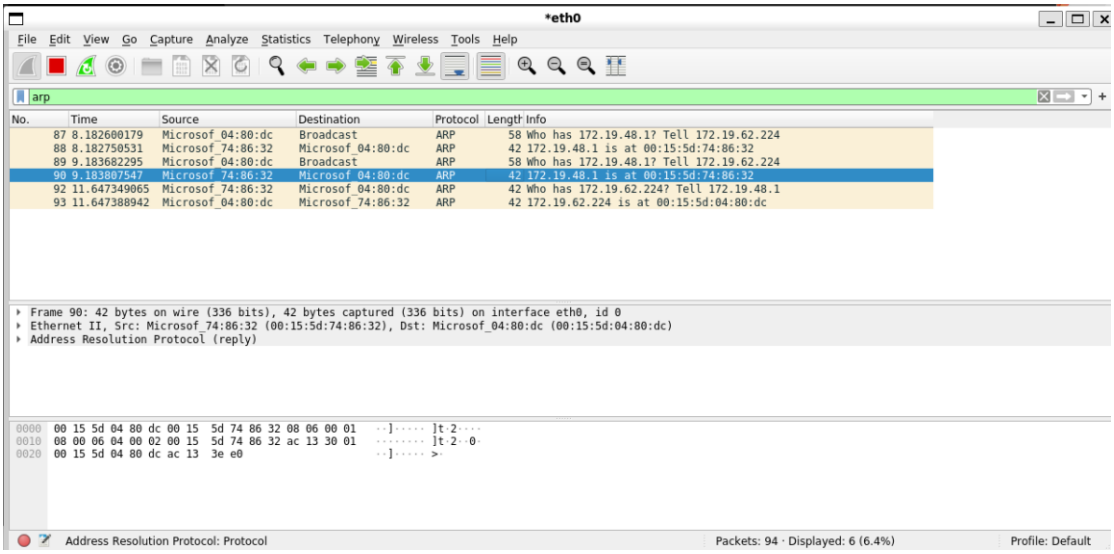
Using UDP requires only **2 packets** (Query + Response).

If a UDP packet is lost, the Operating System simply times out and tries again after a second. This is faster than setting up a TCP connection for every website visit.

Part C: Wireshark and Tshark Packet Capture Analysis

Below is the analysis of the captured network traffic for ARP, ICMP, DNS, UDP, HTTP, and TCP protocols using both the Wireshark GUI and the Tshark command-line tool.

1. ARP (Address Resolution Protocol)




```
root@DESKTOP-3D017KB:/home/zeyun# tshark -i eth0 -f "arp"
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
    1 0.000000000 Microsof_04:80:dc → Broadcast    ARP 58 Who has 172.19.48.1? Tell 172.19.62.224
    2 0.000155700 Microsof_1d:06:c5 → Microsof_04:80:dc ARP 42 172.19.48.1 is at 00:15:5d:1d:06:c5
    3 1.001107412 Microsof_04:80:dc → Broadcast    ARP 58 Who has 172.19.48.1? Tell 172.19.62.224
    4 1.001256431 Microsof_1d:06:c5 → Microsof_04:80:dc ARP 42 172.19.48.1 is at 00:15:5d:1d:06:c5
```

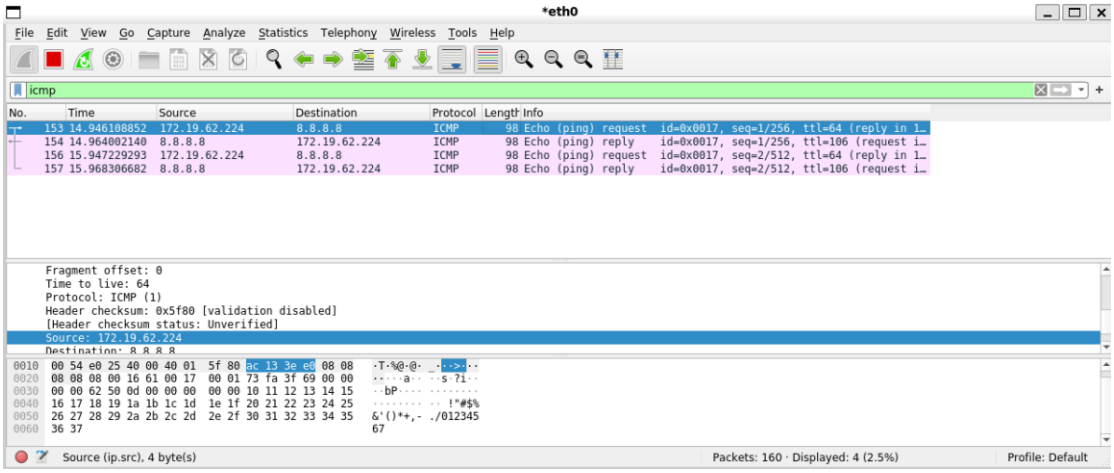
Analysis: The screenshots show the request and reply process of mapping IP addresses

to MAC addresses.

Request: "Who has 172.19.48.1? Tell 172.19.62.224". This is a broadcast message where the sender asks for the physical address of the target.

Reply: "172.19.48.1 is at 00:15:5d:1d:06:c5". The target device responds with its MAC address.

2. ICMP (Internet Control Message Protocol)



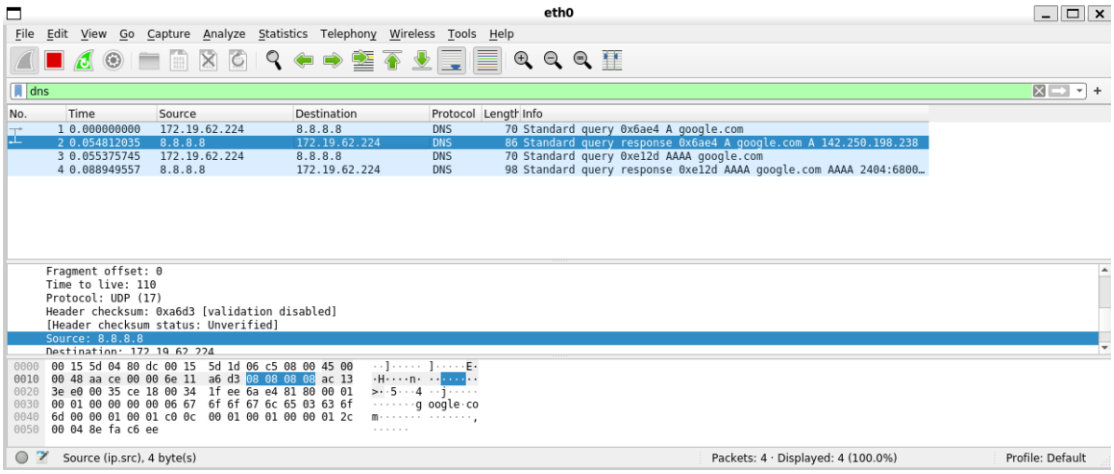

```
root@DESKTOP-3D017KB:/home/zeyun# tshark -i eth0 -f "icmp"
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
    1 0.000000000 172.19.62.224 → 8.8.8.8      ICMP 98 Echo (ping) request  id=0x0018, seq=1/256, ttl=64
    2 0.017238569      8.8.8.8 → 172.19.62.224 ICMP 98 Echo (ping) reply    id=0x0018, seq=1/256, ttl=107 (request in 1)
    3 1.001432359 172.19.62.224 → 8.8.8.8      ICMP 98 Echo (ping) request  id=0x0018, seq=2/512, ttl=64
    4 1.016732340      8.8.8.8 → 172.19.62.224 ICMP 98 Echo (ping) reply    id=0x0018, seq=2/512, ttl=107 (request in 3)
    5 2.002937341 172.19.62.224 → 8.8.8.8      ICMP 98 Echo (ping) request  id=0x0018, seq=3/768, ttl=64
    6 2.018774842      8.8.8.8 → 172.19.62.224 ICMP 98 Echo (ping) reply    id=0x0018, seq=3/768, ttl=107 (request in 5)
```

Analysis: The capture demonstrates a standard ping operation to a Google DNS server (8.8.8.8).

Echo Request (Type 8): Sent from the source to 8.8.8.8.

Echo Reply (Type 0): The server 8.8.8.8 acknowledges the request and sends a reply.

3. DNS (Domain Name System) & UDP




```
root@DESKTOP-3D017KB:/home/zeyun# tshark -i eth0 -f "udp"
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
    1 0.000000000 172.19.62.224 → 8.8.8.8      DNS 70 Standard query 0xeb64 A google.com
    2 0.020393291      8.8.8.8 → 172.19.62.224 DNS 86 Standard query response 0xeb64 A google.com A 142.250.197.46
    3 0.020862268 172.19.62.224 → 8.8.8.8      DNS 70 Standard query 0x4756 AAAA google.com
    4 0.042325819      8.8.8.8 → 172.19.62.224 DNS 98 Standard query response 0x4756 AAAA google.com AAAA 2404:6800:4005
:81f::200e
^C4 packets captured
```

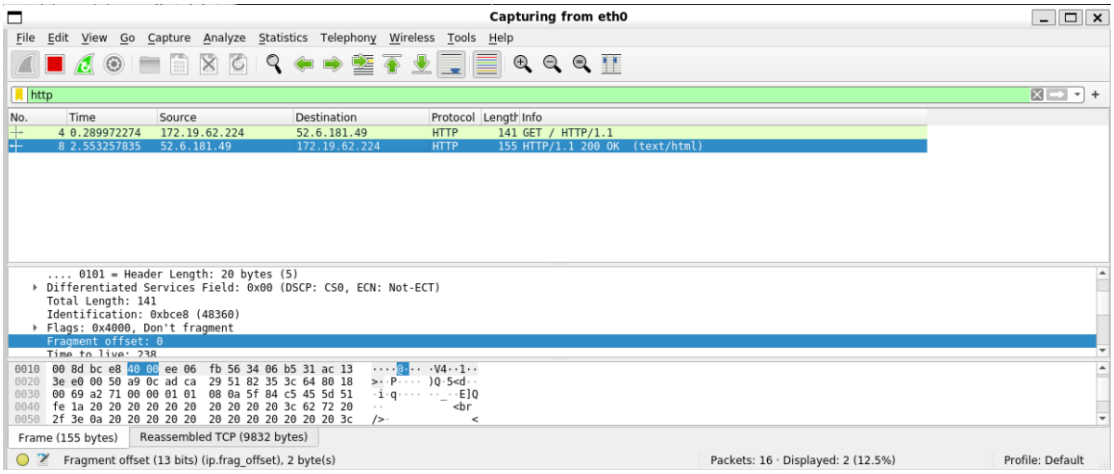
Analysis: DNS traffic runs over UDP (User Datagram Protocol) because it is lightweight and connectionless.

Query: The client sends a standard query (A record) for google.com to the DNS resolver (8.8.8.8).

Response: The resolver returns the IP address 142.250.197.46.

UDP is identified by the lack of handshake (SYN/ACK) flags, unlike TCP.

4. HTTP (HyperText Transfer Protocol) & TCP




```
root@DESKTOP-3D017KB:/home/zeyun# tshark -i eth0 -f "tcp port 80"
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
    1 0.000000000 172.19.62.224 → 18.214.194.42 TCP 74 57570 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval
=2584697148 TSecr=0 WS=128
    2 0.242832774 18.214.194.42 → 172.19.62.224 TCP 74 80 → 57570 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1200 SACK_P
ERM=1 TSval=3627388261 TSecr=2584697148 WS=256
    3 0.242936765 172.19.62.224 → 18.214.194.42 TCP 66 57570 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2584697391 TSe
cr=3627388261
    4 0.243087987 172.19.62.224 → 18.214.194.42 HTTP 141 GET / HTTP/1.1
    5 0.484263938 18.214.194.42 → 172.19.62.224 TCP 66 80 → 57570 [ACK] Seq=1 Ack=76 Win=26880 Len=0 TSval=3627388503 TS
ecr=2584697391
    6 4.352632798 18.214.194.42 → 172.19.62.224 TCP 305 HTTP/1.1 200 OK  [TCP segment of a reassembled PDU]
    7 4.352633239 18.214.194.42 → 172.19.62.224 TCP 9570 80 → 57570 [ACK] Seq=240 Ack=76 Win=26880 Len=9504 TSval=362739
2371 TSecr=2584697391 [TCP segment of a reassembled PDU]
    8 4.352633309 18.214.194.42 → 172.19.62.224 HTTP 155 HTTP/1.1 200 OK  (text/html)
    9 4.352700328 172.19.62.224 → 18.214.194.42 TCP 66 57570 → 80 [ACK] Seq=76 Ack=240 Win=64128 Len=0 TSval=2584701501
TSecr=3627392371
   10 4.352711249 172.19.62.224 → 18.214.194.42 TCP 66 57570 → 80 [ACK] Seq=76 Ack=9744 Win=64640 Len=0 TSval=2584701501
 TSecr=3627392371
   11 4.352715538 172.19.62.224 → 18.214.194.42 TCP 66 57570 → 80 [ACK] Seq=76 Ack=9833 Win=64640 Len=0 TSval=2584701501
 TSecr=3627392371
   12 4.353135324 172.19.62.224 → 18.214.194.42 TCP 66 57570 → 80 [FIN, ACK] Seq=76 Ack=9833 Win=64640 Len=0 TSval=25847
01501 TSecr=3627392371
   13 4.594531846 18.214.194.42 → 172.19.62.224 TCP 66 80 → 57570 [FIN, ACK] Seq=9833 Ack=77 Win=26880 Len=0 TSval=36273
92613 TSecr=2584701501
   14 4.594564219 172.19.62.224 → 18.214.194.42 TCP 66 57570 → 80 [ACK] Seq=77 Ack=9834 Win=64640 Len=0 TSval=2584701743
 TSecr=3627392613
```

Analysis: This capture shows a web request, which relies on the TCP 3-way handshake for reliability.

TCP Handshake:

i. [SYN] sent by the client to initiate connection.
ii. [SYN, ACK] sent by the server to acknowledge.
iii. [ACK] sent by the client to establish the connection.

HTTP Request: GET / HTTP/1.1. The client requests the root resource.

HTTP Response: HTTP/1.1 200 OK. The server successfully returns the requested content (text/html).

D. Trace of Encapsulation and Unencapsulation Processes

This analysis tracks the encapsulation of the HTTP Request (Client to Server) and the unencapsulation of the HTTP Response (Server to Client).

1. Encapsulation Process (HTTP Request)

Direction: Application Layer → Physical Layer **Reference:** Frame 4 (Packet #4 in capture)

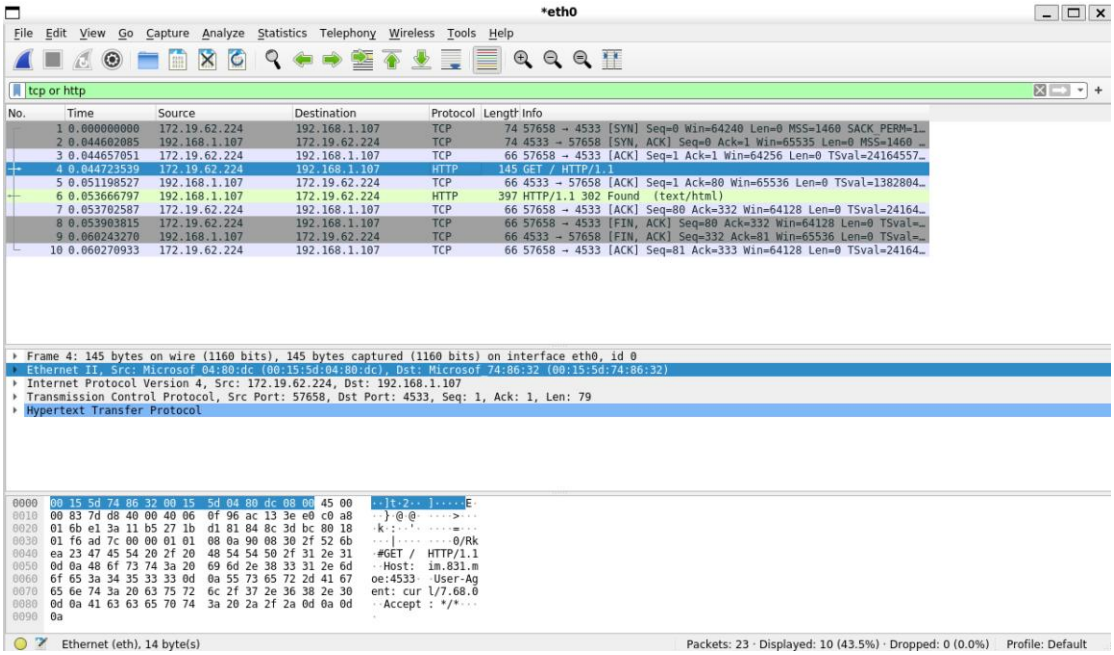
- **Application Layer (L7)**
  - **Action:** The user agent generates an HTTP GET request to access the root path (/).
  - **Core Header Fields:**
    - **Request:** GET / HTTP/1.1
    - **Host:** im.831.moe:4533
    - **User-Agent:** curl/7.68.0
  - **Result:** Data is passed down to the Transport Layer.
- **Transport Layer (L4)**
  - **Action:** The HTTP data is encapsulated into a TCP segment.
  - **Core Header Fields:**
    - **Protocol:** TCP (Transmission Control Protocol)
    - **Source Port:** 57658 (Ephemeral client port)
    - **Destination Port:** 4533 (Server listening port)
    - **Sequence Number:** 1
    - **Len:** 79 (Payload size)
  - **Result:** The TCP segment is passed down to the Network Layer.
- **Network Layer (L3)**
  - **Action:** The TCP segment is encapsulated into an IPv4 packet.
  - **Core Header Fields:**
    - **Protocol:** IPv4
    - **Source IP:** 172.19.62.224 (Client)
    - **Destination IP:** 192.168.1.107 (Server)
    - **Protocol Number:** 6 (TCP)
  - **Result:** The IP packet is passed down to the Data Link Layer.
- **Data Link Layer (L2)**
  - **Action:** The IP packet is encapsulated into an Ethernet II frame.
  - **Core Header Fields:**
    - **Source MAC:** 00:15:5d:74:86:32 (Client Network Interface)
    - **Destination MAC:** 00:15:5d:04:80:dc (Gateway or Next Hop)
    - **Type:** 0x0800 (IPv4)
  - **Result:** The frame is passed to the Physical Layer.
- **Physical Layer (L1)**
  - **Action:** The frame (145 bytes total) is converted into bits and transmitted as electrical or optical signals over the physical medium (Interface eth0).

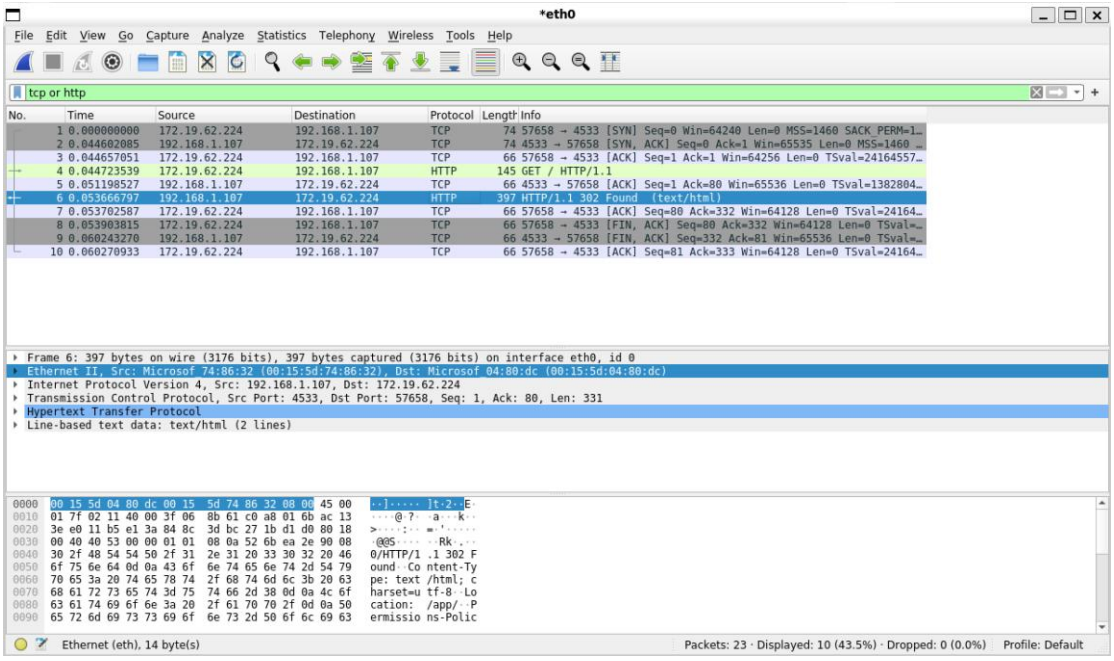
2. Unencapsulation Process (HTTP Response)

Direction: Physical Layer → Application Layer **Reference:** Frame 6 (Packet #6 in capture)

- **Physical Layer (L1)**

    - **Action:** The physical interface (eth0) receives 397 bytes of signals from the wire and converts them into a bitstream for the Data Link Layer.
- **Data Link Layer (L2)**
    - **Action:** The device reads the Ethernet II frame header.
    - **Core Header Fields Checked:**
        - **Destination MAC:** 00:15:5d:74:86:32 (Matches the client's MAC address, so the frame is accepted).
        - **Source MAC:** 00:15:5d:04:80:dc
        - **Type:** 0x0800 (Indicates the payload is IPv4).
    - **Result:** The Ethernet header is stripped, and the payload is passed up to the Network Layer.
- **Network Layer (L3)**
    - **Action:** The device reads the IPv4 header.
    - **Core Header Fields Checked:**
        - **Destination IP:** 172.19.62.224 (Matches the client's IP address).
        - **Source IP:** 192.168.1.107
        - **Protocol:** 6 (Indicates the payload is TCP).
    - **Result:** The IP header is stripped, and the payload is passed up to the Transport Layer.
- **Transport Layer (L4)**
    - **Action:** The device reads the TCP header.
    - **Core Header Fields Checked:**
        - **Destination Port:** 57658 (Matches the open port of the requesting application).
        - **Source Port:** 4533
        - **Sequence Number:** 1
        - **Acknowledgment Number:** 80
    - **Result:** The TCP header is stripped, and the application data is reassembled and passed up to the Application Layer.
- **Application Layer (L7)**
    - **Action:** The HTTP client (browser or script) processes the response data.
    - **Core Header Fields Processed:**
        - **Status Line:** HTTP/1.1 302 Found (Server is redirecting).
        - **Content-Type:** text/html
    - **Result:** The client handles the redirect or displays the content.


*eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp or http
No. Time Source Destination Protocol Length Info
1 0.000000000 172.19.62.224 192.168.1.107 TCP 74 57658 → 4533 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1…
2 0.044602085 192.168.1.107 172.19.62.224 TCP 74 4533 → 57658 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 …
3 0.044657051 172.19.62.224 192.168.1.107 TCP 66 57658 → 4533 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=24164557…
4 0.044723539 172.19.62.224 192.168.1.107 HTTP 145 GET / HTTP/1.1
5 0.051198527 192.168.1.107 172.19.62.224 TCP 66 4533 → 57658 [ACK] Seq=1 Ack=80 Win=65536 Len=0 TSval=1382804…
6 0.053666797 192.168.1.107 172.19.62.224 HTTP 397 HTTP/1.1 302 Found (text/html)
7 0.053702587 172.19.62.224 192.168.1.107 TCP 66 57658 → 4533 [ACK] Seq=80 Ack=332 Win=64128 Len=0 TSval=24164…
8 0.053903815 172.19.62.224 192.168.1.107 TCP 66 57658 → 4533 [FIN, ACK] Seq=80 Ack=332 Win=64128 Len=0 TSval=…
9 0.060243270 192.168.1.107 172.19.62.224 TCP 66 4533 → 57658 [FIN, ACK] Seq=332 Ack=81 Win=65536 Len=0 TSval=…
10 0.060270933 172.19.62.224 192.168.1.107 TCP 66 57658 → 4533 [ACK] Seq=81 Ack=333 Win=64128 Len=0 TSval=24164…
Frame 4: 145 bytes on wire (1160 bits), 145 bytes captured (1160 bits) on interface eth0, id 0
Ethernet II, Src: Microsof 04:80:dc (00:15:5d:04:80:dc), Dst: Microsof 74:86:32 (00:15:5d:74:86:32)
Internet Protocol Version 4, Src: 172.19.62.224, Dst: 192.168.1.107
Transmission Control Protocol, Src Port: 57658, Dst Port: 4533, Seq: 1, Ack: 1, Len: 79
Hypertext Transfer Protocol
0000 00 15 5d 74 86 32 00 15 5d 04 80 dc 08 00 45 00
0010 00 83 7d d8 40 00 40 06 0f 96 ac 13 3e e0 c0 a8
0020 01 6b e1 3a 11 b5 27 1b d1 81 84 8c 3d bc 80 18
0030 01 f6 ad 7c 00 00 01 01 08 0a 90 08 30 2f 52 6b
0040 ea 23 47 45 54 20 2f 20 48 54 54 50 2f 31 2e 31
0050 0d 0a 48 6f 73 74 3a 20 69 6d 2e 38 33 31 2e 6d
0060 6f 65 3a 34 35 33 33 0d 0a 55 73 65 72 2d 41 67
0070 65 6e 74 3a 20 63 75 72 6c 2f 37 2e 36 38 2e 30
0080 0d 0a 41 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 0d
0090 0a
Ethernet (eth), 14 byte(s)
Packets: 23 · Displayed: 10 (43.5%) · Dropped: 0 (0.0%)
Profile: Default


*eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp or http
No. Time Source Destination Protocol Length Info
1 0.000000000 172.19.62.224 192.168.1.107 TCP 74 57658 → 4533 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1…
2 0.044602085 192.168.1.107 172.19.62.224 TCP 74 4533 → 57658 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 …
3 0.044657051 172.19.62.224 192.168.1.107 TCP 66 57658 → 4533 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=24164557…
4 0.044723539 172.19.62.224 192.168.1.107 HTTP 145 GET / HTTP/1.1
5 0.051198527 192.168.1.107 172.19.62.224 TCP 66 4533 → 57658 [ACK] Seq=1 Ack=80 Win=65536 Len=0 TSval=1382804…
6 0.053666797 192.168.1.107 172.19.62.224 HTTP 397 HTTP/1.1 302 Found (text/html)
7 0.053702587 172.19.62.224 192.168.1.107 TCP 66 57658 → 4533 [ACK] Seq=80 Ack=332 Win=64128 Len=0 TSval=24164…
8 0.053903815 172.19.62.224 192.168.1.107 TCP 66 57658 → 4533 [FIN, ACK] Seq=80 Ack=332 Win=64128 Len=0 TSval=…
9 0.060243270 192.168.1.107 172.19.62.224 TCP 66 4533 → 57658 [FIN, ACK] Seq=332 Ack=81 Win=65536 Len=0 TSval=…
10 0.060270933 172.19.62.224 192.168.1.107 TCP 66 57658 → 4533 [ACK] Seq=81 Ack=333 Win=64128 Len=0 TSval=24164…
Frame 6: 397 bytes on wire (3176 bits), 397 bytes captured (3176 bits) on interface eth0, id 0
Ethernet II, Src: Microsof 74:86:32 (00:15:5d:74:86:32), Dst: Microsof 04:80:dc (00:15:5d:04:80:dc)
Internet Protocol Version 4, Src: 192.168.1.107, Dst: 172.19.62.224
Transmission Control Protocol, Src Port: 4533, Dst Port: 57658, Seq: 1, Ack: 80, Len: 331
Hypertext Transfer Protocol
Line-based text data: text/html (2 lines)
0000 00 15 5d 04 80 dc 00 15 5d 74 86 32 08 00 45 00
0010 01 7f 02 11 40 00 3f 06 8b 61 c0 a8 01 6b ac 13
0020 3e e0 11 b5 e1 3a 84 8c 3d bc 27 1b d1 d0 80 18
0030 00 40 40 53 00 00 01 01 08 0a 52 6b ea 2e 90 08
0040 30 2f 48 54 54 50 2f 31 2e 31 20 33 30 32 20 46
0050 6f 75 6e 64 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79
0060 70 65 3a 20 74 65 78 74 2f 68 74 6d 6c 3b 20 63
0070 68 61 72 73 65 74 3d 75 74 66 2d 38 0d 0a 4c 6f
0080 63 61 74 69 6f 6e 3a 20 2f 61 70 70 2f 0d 0a 50
0090 65 72 6d 69 73 73 69 6f 6e 73 2d 50 6f 6c 69 63
Ethernet (eth), 14 byte(s)
Packets: 23 · Displayed: 10 (43.5%) · Dropped: 0 (0.0%)
Profile: Default